Steroid Level Detection in Milk Production

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- ***Abstract***

Steroid hormone contamination in milk has become a growing concern due to its potential endocrine-disrupting effects on human health. These compounds originate from both natural physiological processes in dairy animals and the misuse of veterinary drugs. This paper presents a comprehensive and updated review of steroid detection techniques in milk production, focusing on developments from 2021 to 2026. Advanced analytical tools such as ultra-high-performance liquid chromatography coupled with tandem mass spectrometry (UHPLC-MS/MS), gas chromatography–mass spectrometry (GC-MS), and emerging extraction techniques are discussed in detail. Recent studies demonstrate the ability to simultaneously detect up to 58 steroid residues with high sensitivity and accuracy. (ScienceDirect)

This paper also explores detection workflows, challenges, regulatory standards, and future trends such as biosensors and AIdriven monitoring systems. The integration of efficient extraction methods and automated analytical tools is significantly improving food safety monitoring systems worldwide.

- ***Keywords***

Steroid hormones, Milk safety, UHPLC-MS/MS, LC-MS, Dairy contamination, Food safety monitoring

- ***I. Introduction***

Milk is one of the most widely consumed food products globally. However, the presence of steroid residues in milk poses significant health risks. These compounds include:

- ***Natural hormones (estradiol, progesterone)***
- ***Synthetic growth promoters***
- ***Veterinary drug residues***

Steroid contamination may occur due to:

- ***Hormonal treatments in dairy cattle***
- ***Environmental contamination***
- ***Improper veterinary practices***

Recent research shows that even trace levels of steroid residues can interfere with human endocrine systems, particularly in children and vulnerable populations.

- ***II. Classification of Steroids in Milk***
- ***A. Natural Steroids***

Produced biologically in animals:

- ***Estrogens***
- ***Androgens***
- ***Progestogens***
- ***B. Synthetic Steroids Artificially introduced:***
- ***Trenbolone***
- ***Diethylstilbestrol***
- ***C. Glucocorticoids***

Used in veterinary medicine:

- ***Cortisol***
- ***Cortisone***

Recent findings detected hydrocortisone and cortisone in commercial milk samples, though within safe limits. (ScienceDirect)

- ***Pie Chart Representation – Steroid Distribution***

[Pie Chart – Approximate Distribution in Milk Samples]

Natural Steroids : 45%

Glucocorticoids : 30%

Synthetic Steroids : 15%

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Mycotoxin-derived : 10%

- *III. Sources of Steroid Contamination*
- *1. Animal Metabolism*

Naturally secreted hormones enter milk.

- *2. Veterinary Drug Use*
- *Growth promotion*
- *Disease treatment*
- *3. Environmental Exposure*
- *Contaminated feed*
- *Water sources*

- *IV. Analytical Techniques for Detection*
- *A. UHPLC-MS/MS (Gold Standard) The most advanced method currently used.*
- *Key Features:*
- *Detects multiple steroids simultaneously*
- *Extremely low detection limits (ng/L) Recent study:*
- *Detection of 58 glucocorticoids*
- *Detection limits: 0.1–1.0 µg/kg*
- *High accuracy ($R^2 > 0.99$) (ScienceDirect)*

- *Flowchart: UHPLC-MS/MS Detection Process*

Milk Sample Collection

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Pre-treatment (Centrifugation)

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Extraction (SPE / QuEChERS)

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Chromatographic Separation

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Mass Spectrometry Detection

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Data Analysis & Quantification

- *B. LC-MS/MS*

Widely used alternative:

- *Detects multiple hormone classes*
- *High sensitivity*

- *C. GC-MS*

Traditional but reliable:

- *Suitable for volatile steroid compounds*
- *Requires derivatization*

- *D. Emerging Techniques*
- *1. QuEChERS Method*
- *Fast and cost-effective*
- *Used with UPLC-MS/MS*
- *Detects multiple steroids simultaneously (RSC publication)*
- *2. High-Resolution Mass Spectrometry (HRMS)*
- *Detects unknown metabolites*
- *Very high precision*
- *3. Liquid-Liquid Extraction (LLE)*
- *Effective for complex matrices*
- *Used in modern workflows (ScienceDirect)*

- *V. Comparison of Detection Techniques Sensitivity,Accuracy, Speed and Cost.*

UHPLC-MS/MS offers **very high sensitivity, excellent accuracy, fast speed**, but comes with a high cost.

LC-MS/MS provides **moderate sensitivity, medium accuracy, high speed**, and has a good (moderate) cost.

GC-MS HRMS delivers **high sensitivity, good accuracy, very fast speed**, but incurs a very high cost.

- *VI. Sample Preparation Techniques*

Milk is a complex biological matrix containing:

- *Proteins*
- *Fats*
- *Sugars*

- *Common Preparation Methods:*
- *Solid Phase Extraction (SPE)*
- *QuEChERS*
- *Liquid-liquid extraction*

- *Diagram: Sample Preparation Workflow*

Raw Milk Sample

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Protein Removal

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Fat Separation

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Extraction (SPE / LLE)

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Clean-up Stage

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Ready for Analysis

- *VII. Challenges in Steroid Detection*
- *1. Matrix Complexity*

Milk components interfere with detection.

- *2. Ultra-Low Concentrations Steroid present at*

trace levels

- *3. Isomer Similarity*

Structural similarity makes separation difficult.

- *4. Regulatory Compliance*

Strict detection limits required.

- *VIII. Recent Innovations (2021–2026)*
- *1. Multi-Residue Detection*
- *Detection of up to 58 steroids simultaneously (ScienceDirect)*
- *2. AI-Based Data Analysis*
- *Improved peak identification*
- *Reduced human error*
- *3. Green Chemistry Approaches*
- *Reduced solvent use*
- *Faster processing*
- *4. High Sensitivity Detection*

Modern methods achieve pg/mL level sensitivity (ScienceDirect)

- *IX. Applications in Dairy Industry*
- *Milk quality control*
- *Export certification*
- *Veterinary drug monitoring*
- *Consumer safety assurance*

- *X. Case Study (Recent Research)*

A 2024 study developed a UHPLC-MS/MS method:

- *Simultaneous detection of 58 glucocorticoids*
- *Recovery: 71–113%*
- *Precision: <15% deviation (ScienceDirect)*

This demonstrates high reliability for real-world milk testing.

- *XI. Future Scope*
- *1. Biosensors*
- *3. AI Integration*
- *Automated contamination prediction*
- *4. IoT-Based Dairy Monitoring*
- *Continuous quality tracking*

- *Flowchart: Future Smart Dairy Monitoring System*

Milk Production Unit

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IoT Sensors

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Real-time Steroid Detection

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Cloud Data Analysis (AI)

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Alert System

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Regulatory Compliance

- *XII. Ethical and Regulatory Considerations*
- *Strict adherence to Maximum Residue Limits (MRLs)*
- *Monitoring misuse of veterinary drugs*
- *Ensuring public health safety*

- *XIII. Discussion*

Modern analytical methods have significantly improved detection capabilities. However:

- *High equipment cost limits accessibility*
- *Skilled professionals required*
- *Need for portable and affordable solutions*

- *Where:*
- *CCC = concentration of steroid in milk*
- *AsA_sAs = peak area of sample*
- *AstdA_{std}Astd = peak area of standard*
- *CstdC_{std}Cstd = known concentration of standard*

Calibration Curve Equation y=mx+cy = mx + cy=mx+c

yyy = detector response

- *xxx = concentration*
- *mmm = slope*
- *ccc = intercept*

II. Limit of Detection (LOD)

LOD=3σmLOD = \frac{3\sigma}{m}LOD=m3σ

- *Where:*
- *σ\sigmaσ = standard deviation*
- *mmm = slope of calibration curve*

- *XIV. Mathematical Model for Steroid Level Calculation*
- *1. Portable Detection Devices*
- *2. Concentration Formula*
- *On-farm analysis*
- *C=AsAstd×CstdC = \frac{A_s}{A_{std}} \times C_{std}C=Astd levels (ng/L).*

XV. Conclusion

Steroid detection in milk production has evolved rapidly due to advancements in chromatographic and mass spectrometric techniques. UHPLC-MS/MS remains the most reliable method, offering high sensitivity and multi-residue detection. Future developments will likely focus on real-time monitoring and AI integration, ensuring safer dairy products globally.

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XVI. References

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